Østfold University College

A Systematic Literature Review of DevOps in quantum software engineering

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Kadir Amini, Berat Murseli

Avdeling for Informasjonsteknologi
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Abstract

Acknowledgments

Contents

Abstract **i**

Acknowledgments **iii**

Contents **v**

List of Figures **vii**

List of Tables **ix**

1 Introduction **1**
1.1 Goal 1
1.2 Background 1
1.3 Research Questions 2

2 Literature Review **3**

3 Methodology **5**
3.1 Review Design 5
3.2 PICOC framing 5
3.3 Search strategy 5
3.4 Inclusion and exclusion criteria 6
3.5 Study selection process 7
3.6 Quality assessment 7
3.7 Data extraction plan 8
3.8 Data synthesis plan 9

4 Results **11**
4.1 Overview of Included Studies 11
4.2 RQ1: DevOps Practices in Quantum Software Enginee 11
4.3 RQ2: CI/CD and Automation Challenges 11
4.4 RQ3: Barriers to Adoption 11

5 Discussion **13**
5.1 Summary 13

6 Conclusion **15**

Bibliography **17**

List of Figures

List of Tables

3.1 PICOC table 6
3.2 Inclusion and Exclusion Criteria Table 7
3.3 Quality assessment questions and scoring 8
3.4 Placeholder Caption 9

Chapter 1

Introduction

Quantum computing has developed rapidly in recent years, with cloud-accessible quantum processors now available from IBM, Google, Amazon, and Microsoft. However, the software engineering side of the field is still relatively immature. Much of the existing research focuses on algorithms, hardware, and performance, while less attention has been given to development workflows, testing pipelines, deployment practices, and automation in quantum software projects.

DevOps has quickly become a dominant part in software engineers ability to deliver software quickly and reliably. continuous integration (CI), automated testing, containerized deployment, and more are standard practices for software development. If quantum software is ever to go beyond science research and become deployable technology, the ability to deliver, maintain, and develop it will require DevOps.

quantum computers are different from classical computers at a fundamental level. Quantum computer are probabilistic due to relying on quantum mechanics to represent the binary values. Quantum computers are sensitive to noise, qubit calibration, decoherence, gate errors, SDK verion, and more. All this results in the same program potentially behaving differently on when and where it is executed.

1.1 Goal

A key objective of this SLR is to identify research gaps in the field of quantum computing, more specifically within DevOps and CI/CD. By identifying gaps in the existing research, this paper aims to provide direction for future studies and contribute to helping quantum computing researchers move closer to practical solutions in this field.

A second objective is to provide a theoretical framework for how DevOps and CI/CD practices can be integrated into the field of quantum computing. This aims to provide an understanding of how development, testing, deployment, and continuous integration processes can be adapted to make use of the strengths of quantum software development. By doing so, this paper contributes ideas on how quantum computing can reduce the gap between concept and reality.

1.2 Background

The purpose of this project is to gain a better understanding of how quantum computing works. With the advanced technology available today, we wanted to explore how modern computing technology is built, what principles lie behind the development of quantum computing, and how this emerging field may influence future technological development.

Understanding the concept of quantum computing is a large field, and it would take too much time to cover everything. Therefore, we chose to focus more on the field of DevOps and CI/CD, especially understanding the fundamentals behind the integration of quantum computing. We wanted to explore how existing applications are adapting to and using this technology today, and whether practical use cases already exist.

Spørsmål til lører: trenger vi info om quantom software i bunn?

1.3 Research Questions

This review is guided by the following research questions:

- **RQ1:** How does quantum computing DevOps compare to standard DevOps?
- **RQ2:** What challenges exist in implementing continuous integration and continuous delivery for quantum software?
- **RQ3:** What are the main barriers preventing the adoption of standard DevOps practices in quantum software engineering?

These questions allow the review to move beyond a simple mapping of papers. They make it possible to compare quantum software workflows with classical DevOps, identify technical and organizational challenges, and explain why adoption is currently limited.

Chapter 2

Literature Review

Before initiating the SLR we find it greatly important to explore the landscape of the current state of research within quantum computing. The landscape of quantum software engineering is rapidly advancing as such taking a quick peek into the literature will help us understand

[1]

[2]

write if a slr has been done on the topic.

other related literature

Chapter 3

Methodology

snakke om kitchanham som en introduction

3.1 Review Design

This project follows a systematic literature review methodology consisting of planning to get better control of the structure, then conducting and reporting phases to get an overview of the existing research. This follows Kitchenham's [3] guidelines, which emphasize a transparent, systematic, and structured review process to reduce bias.

The planning phase established the research questions, the search string, the inclusion and exclusion criteria, the quality-assessment instrument, and the data-extraction template. The conducting phase executed searches on the six databases as chosen, applied the criteria in two screening stages for the first initial search, and ran one round of backward snowballing on the most-cited studies. Then, three stages were carried out, adding quality assessment and scoring, and then extracting data synthesis.

3.2 PICOC framing

PICOC stands for Population, Intervention, Comparison, Outcome, and Context. These elements provide structure and help define what the review is about, what type of information we are trying to gather, and what we want to compare based on the research topic.

In our review, we chose to use these elements to help justify the research focus and identify the most important information based on the available papers. This helps improve the quality of the review and ensures that the selected studies are relevant for future work.

This is our PICOC table:

3.3 Search strategy

To cover the topics of quantum software and quantum computing, we selected several academic databases for our systematic literature review. These databases were chosen because they provide access to high-quality, peer-reviewed research relevant to our study.

- **ACM**

Element	Definition for this review
Population	Quantum software projects, quantum programs, developers, and teams using quantum SDKs or quantum development workflows.
Intervention	DevOps practices, CI/CD, workflow automation, testing pipelines, version control, reproducibility, and related engineering methods.
Comparison	Standard DevOps or classical software engineering practices when a comparison is possible.
Outcome	Productivity, reliability, maintainability, automation level, correctness, reproducibility, and identified barriers.
Context	Quantum software engineering in academia or industry, including simulator-based and hardware-based settings.

Table 3.1: PICOC table

- **IEEE Xplore**
- **Science Direct**
- **Springer Link**
- **Wiley**
- **Taylor And Francis**

We started with Google Scholar and tested different search strings. Several of the first search strings returned a very large number of papers because they were too broad. Therefore, we made the search string more specific and tested different versions until we found a more complete search string that returned around 4,000 papers. This gave us a better basis for finding relevant and high-quality papers. After that, we used the chosen search string to search in the different databases we had selected.

Our selected search string is: **("quantum software" OR "quantum program" OR "quantum programming" OR "quantum computing") AND ("DevOps" OR "CI/CD")**

3.4 Inclusion and exclusion criteria

As part of our review method, we needed to define criteria for what should be included and excluded in our SLR. Since our language knowledge is mainly based on English, we chose to include only papers written in English and exclude papers written in other languages. This makes it possible for us to read the papers and understand the information.

We also chose to include only peer-reviewed papers, as this helps ensure a higher level of quality. Papers that did not meet the required quality level were excluded.

There is a large amount of research on quantum computing, and some papers focus mainly on quantum physics. These papers were excluded because our review focuses specifically on quantum software development and software engineering.

We wanted to focus on newer research in the field, so we excluded papers published before 2020. This helps ensure that the review includes more recent developments and innovations.

To make our research more relevant to the search field, we also excluded papers that did not focus on topics such as DevOps, CI/CD, workflow automation, and testing pipelines. This helped us narrow the review and identify papers that were more closely related to our research topic.

Inclusion criteria	**Exclusion criteria**
Written in English	Non-English publications
Peer-reviewed articles or clearly relevant scholarly papers	Non-peer-reviewed material with no academic value
Focuses on quantum software development or quantum software engineering	Pure quantum physics or hardware papers with no software relevance
Published between 2020 and 2026	Published before 2020 unless needed for background only
Discusses DevOps, CI/CD, workflow automation, testing pipelines, or closely related practices in a quantum context	No meaningful connection to DevOps or software-engineering practice in a quantum context

Table 3.2: Inclusion and Exclusion Criteria Table

3.5 Study selection process

For the study selection process, we started with title and abstract screening to remove irrelevant studies. This means that we went through all the papers we found and reviewed their titles and abstracts to remove studies that were not related to our topic. After the first stage, we conducted full-text screening using inclusion and exclusion criteria to make the selection more relevant. Then, we carried out a quality assessment to ensure that the chosen papers were of high quality and suitable for our research. In the final stage, we performed data extraction and synthesis as the end result. This process follows the guidelines by [3]

3.6 Quality assessment

Each research paper that passed full-text screening was then quality assessed with a given score based on the quality assessment questions, which were partly implemented from Kitchenham's QA checklist [3]. As this study is a qualitative paper, we also adopted some of the QA questions from the qualitative study checklist.

QA #	Question	Scoring
QA1	Are the research aims clearly stated and relevant to quantum software engineering, DevOps, CI/CD, testing, or automation?	Yes (1) / Partly (0.5) / No (0)
QA2	How credible is the paper?	Yes / Partly / No
QA3	How well is has diversity off perspective and context been explored in the paper?	Yes / Partly / No
QA4	Does the study include validation, prototype, experiment, simulation, CI/CD pipeline	Yes / Partly / No
QA5	Does the study discuss limitations, challenges, or barriers to adoption?	Yes / Partly / No

Table 3.3: Quality assessment questions and scoring

The maximum score for each study was 5. Each quality assessment question was scored as Yes = 1, Partly = 0.5, and No = 0. The scores were used to give an overall quality of the selected papers and to support the final selection of studies. Studies with lower scores were considered less reliable and were not prioritized in the synthesis.

3.7 Data extraction plan

The purpose of the data extraction plan is to ensure that the same information is collected from each paper and that the extraction process is consistent for all selected studies. This helps ensure that relevant information is extracted while reducing the risk of missing important details.

In the extraction fields, we included author, year, venue, paper type, and database source. These fields help ensure that the extracted information is relevant to the research questions of the review. They also make it easier to describe trends among the papers, such as publication year, which was important for assessing the relevance of each paper.

The second field was included to ensure that the extracted information was related to the main topic of the research. This included papers that discuss workflow, testing, automation, deployment, version control, and CI/CD. This field was highly important because it helped determine whether each paper was relevant to the core focus of the review.

The contribution type was used to classify whether the paper presented a tool, framework, method, benchmark, or review. This helped us understand the type of contribution each paper made and compare the papers more systematically.

To further assess the quality of the papers, the evidence type was also extracted. This showed how the claims in each paper were supported. The evidence types included experiments, case studies, benchmarks, and surveys. This field was useful for identifying whether the paper was practical, empirical, or conceptual.

The main findings were key result related to research questions 1, 2, and 3. This ensured that the selected papers were aligned with the scope of the review and made it easier to identify similarities and differences between the papers.

We also included challenges and barriers, such as technical, organizational, tooling, reproducibility, and skill-related barriers. In addition, a quality score was added based on the quality checklist. This helped strengthen the assessment of each paper and made the review more reliable.

Overall, the data extraction plan provided a clear and structured way to collect and organize information from the selected papers. It helped ensure that the extracted data was strongly related to the research questions and supported a systematic analysis of the literature.

Field	**Purpose**
Paper metadata	Authors, year, venue, paper type, and database source.
Quantum software focus	Whether the paper discusses workflow, testing, automation, deployment, version control, or CI/CD.
Contribution type	Tool, framework, method, benchmark, review.
Evidence type	Experiment, case study, benchmark, survey, position paper.
Main findings	Key results related to RQ1, RQ2, or RQ3.
Barriers/challenges	Technical, organizational, tooling, reproducibility, hardware-access, or skills-related barriers.
Quality score	Assessment result from the quality checklist.

Table 3.4: Placeholder Caption

3.8 Data synthesis plan

after the data was extracted we

Chapter 4

Results

4.1 Overview of Included Studies

4.2 RQ1: DevOps Practices in Quantum Software Enginee

4.3 RQ2: CI/CD and Automation Challenges

4.4 RQ3: Barriers to Adoption

Chapter 5

Discussion

5.1 Summary

Chapter 6

Conclusion

Bibliography

[1] M. A. Akbar, S. Rafi, A. Khan, and B. Ye, "Quantum secure devops (qsecops): Integrating quantum-based security checks into ci/cd pipelines: Next-generation software security," in *Proceedings of the 33rd ACM International Conference on the Foundations of Software Engineering*, FSE Companion '25, (New York, NY, USA), p. 1732–1741, Association for Computing Machinery, 2025.

[2] A. AlSanad and M. Akbar, "Optimizing devops enablers for quantum software development." Preprint, Research Square, Nov. 2023. Version 1, posted 14 November 2023. Not peer reviewed.

[3] B. Kitchenham and S. Charters, "Guidelines for performing systematic literature reviews in software engineering," Technical Report EBSE-2007-01, EBSE, 2007.